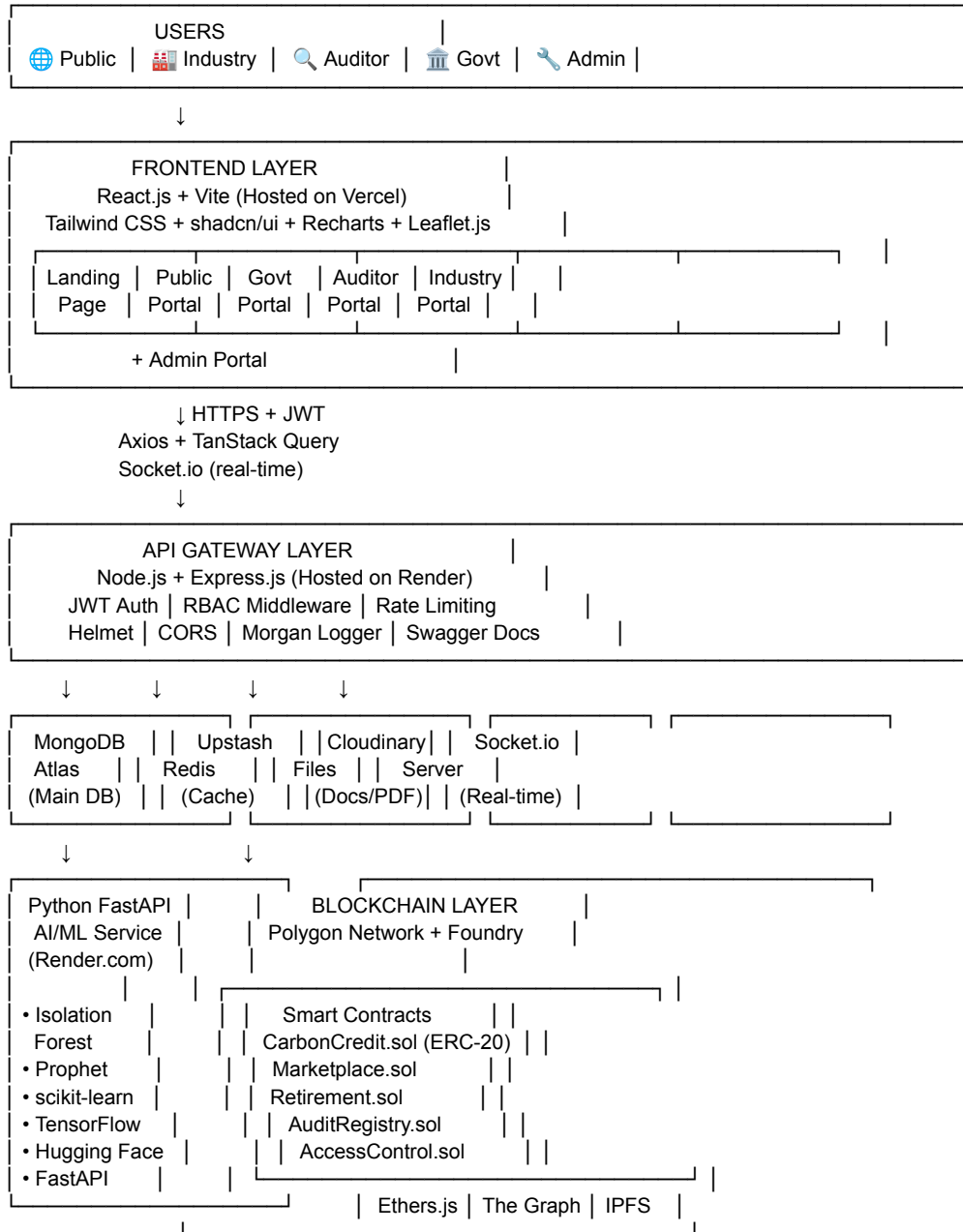




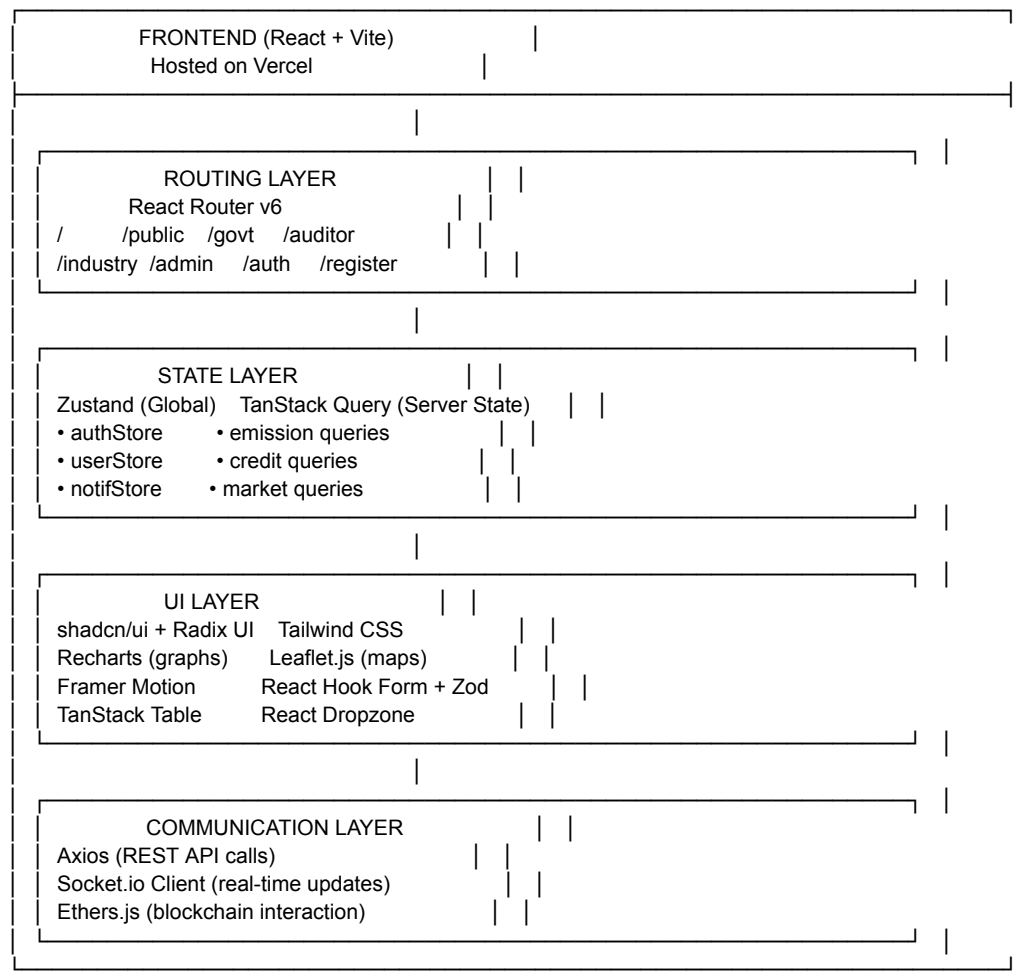
Complete System Architecture

1 High-Level Architecture Overview

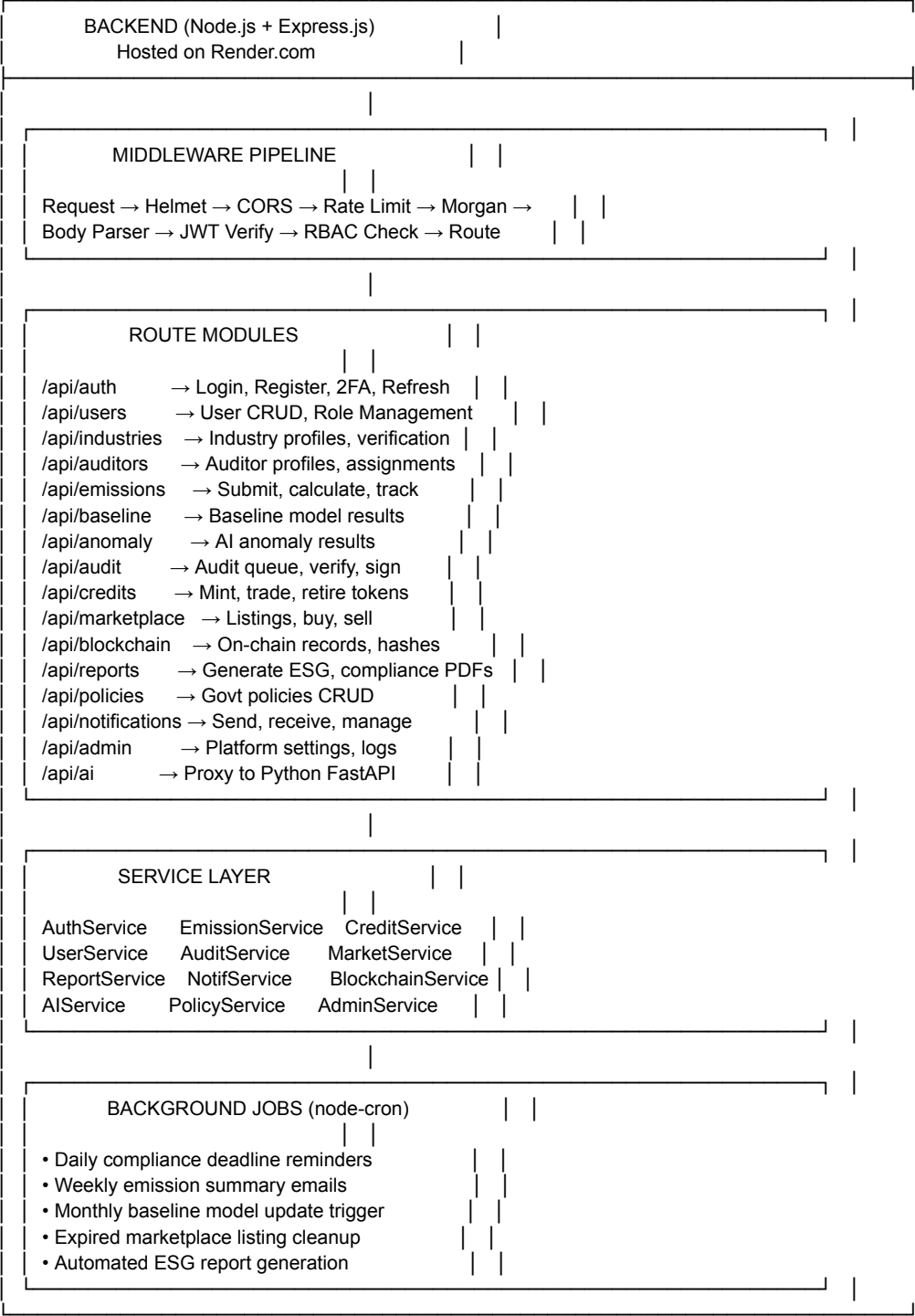


2 Detailed Layer Architecture

🍷 Frontend Layer



⚙️ Backend Layer



Database Layer

DATABASE LAYER			
MongoDB Atlas (Primary DB)	Upstash Redis (Cache)	Cloudinary (File Storage)	
Collections: • users • industries • auditors • governments • submissions • baselines • anomaly_reports • audit_reports • carbon_credits • transactions • marketplace • penalties • policies • notifications • activity_logs • emission_factors Free Limit: 512MB	Cache Keys: • session:userId • otp:email • rateLimit:ip • emissions:id • baseline:indId • market:list • notif:userId TTL Settings: • OTP: 5 mins • Session: 24hrs • Rate: 1 min • Cache: 10 mins Free: 10k/day	Folders: • /documents (registration docs) • /reports (ESG PDFs) • /certificates (compliance certs) • /policies (govt docs) • /avatars (profile photos) Free Limit: 25GB	

AI / ML SERVICE (Python FastAPI)
Hosted on Render.com

API ENDPOINTS

POST /ai/anomaly-detect → Check submission
POST /ai/baseline-model → Calculate baseline
POST /ai/predict-emissions → Forecast next period
POST /ai/suggestions → Reduction recommendations
POST /ai/whatif-simulate → What-if CO₂ calculator
POST /ai/fraud-score → Industry fraud risk
POST /ai/generate-report → NLG ESG report text
GET /ai/national-forecast → Country emission trend
GET /ai/credit-price → Price prediction

ML MODELS

Model 1: Anomaly Detection
→ Algorithm: Isolation Forest
→ Input: emission data per source
→ Output: anomaly flag + risk score (0-100)

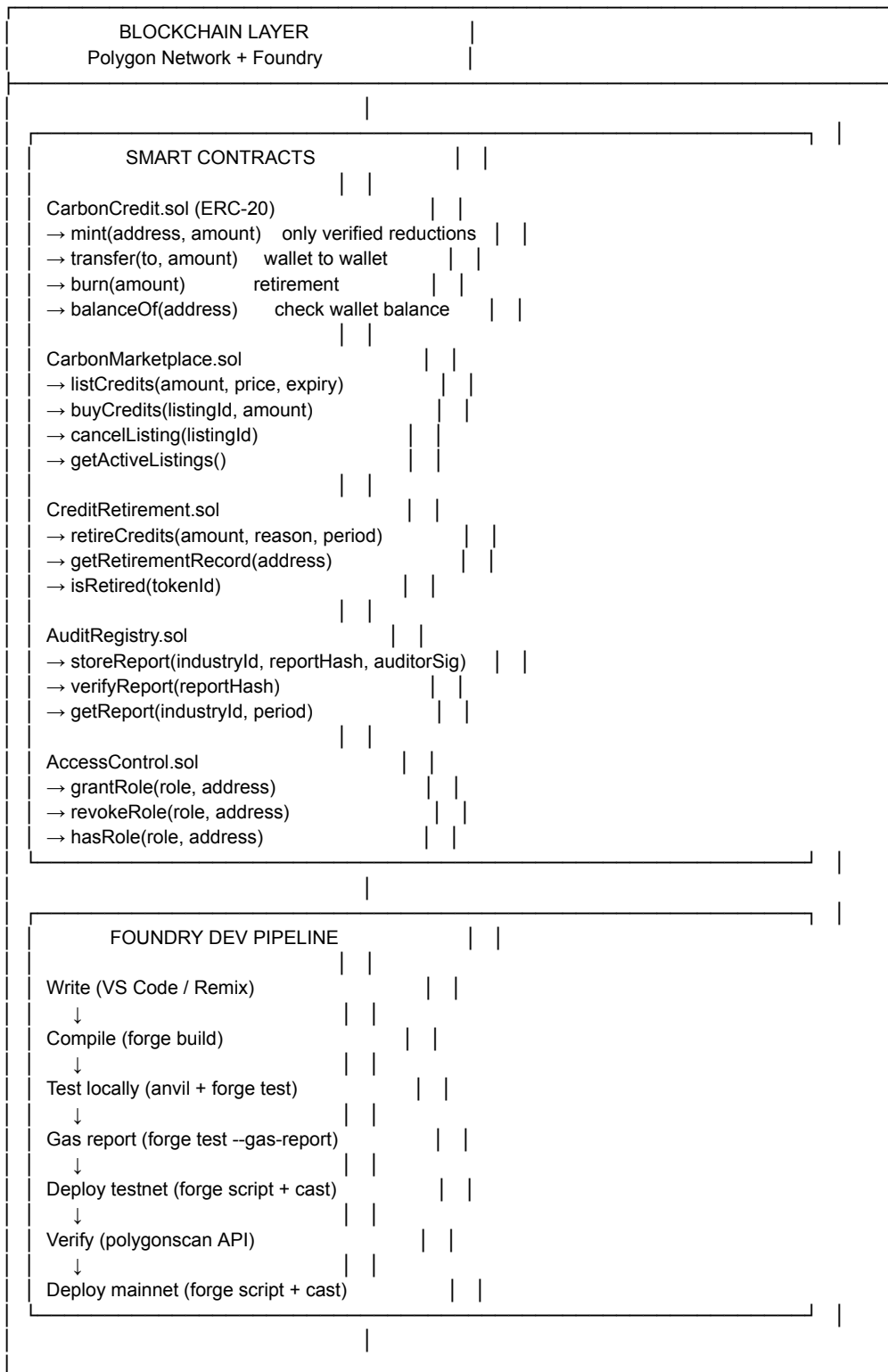
Model 2: Baseline Calculator
→ Algorithm: Linear Regression + K-Means
→ Input: historical emissions + sector data
→ Output: expected baseline (tCO₂e) + range

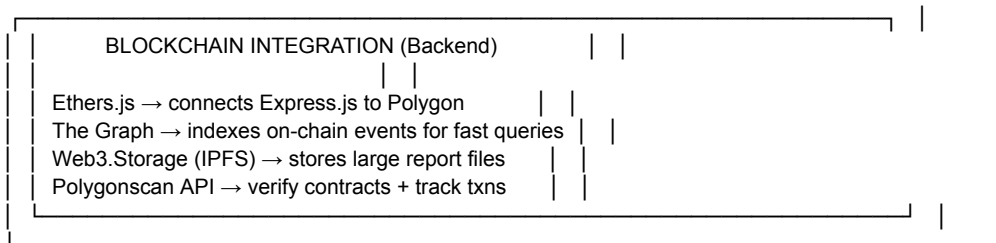
Model 3: Emission Forecasting
→ Algorithm: Prophet + LSTM (TensorFlow)
→ Input: time-series emission data
→ Output: next period forecast + confidence band

Model 4: Fraud Detection
→ Algorithm: One-Class SVM
→ Input: submission patterns + history
→ Output: fraud probability score

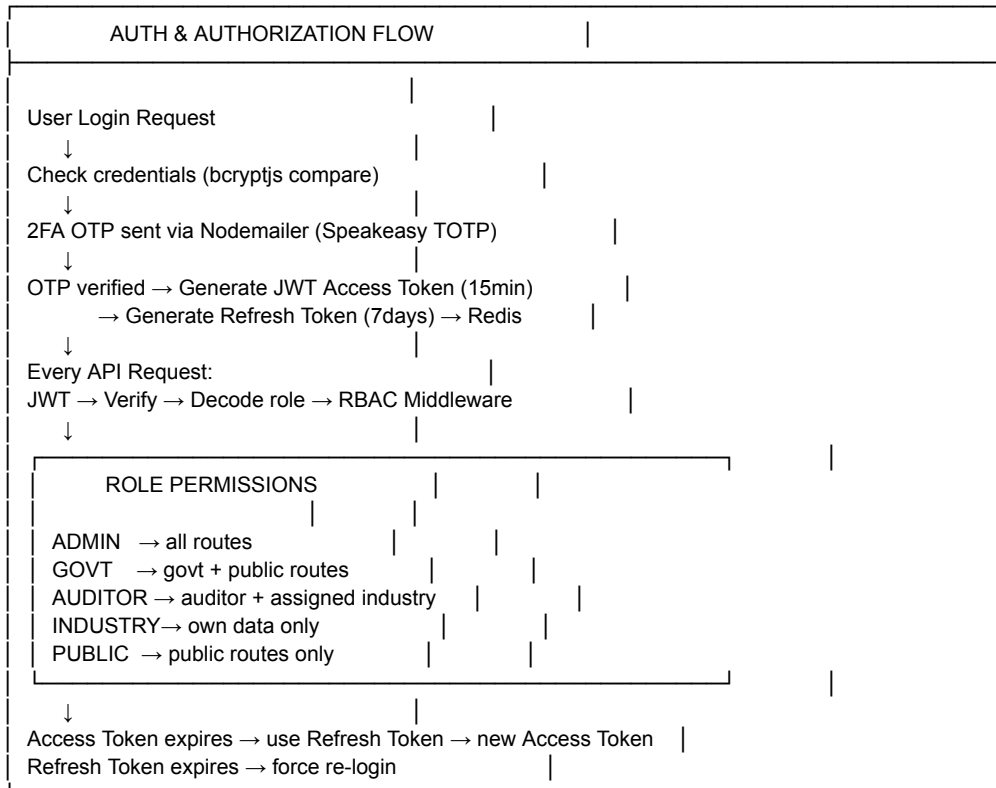
Model 5: Reduction Suggestions
→ Algorithm: Feature Importance + Rule Engine
→ Input: emission breakdown by source
→ Output: ranked actionable suggestions

Model 6: Credit Price Prediction
→ Algorithm: Prophet time-series
→ Input: historical credit prices + trade volume
→ Output: price forecast + trend direction

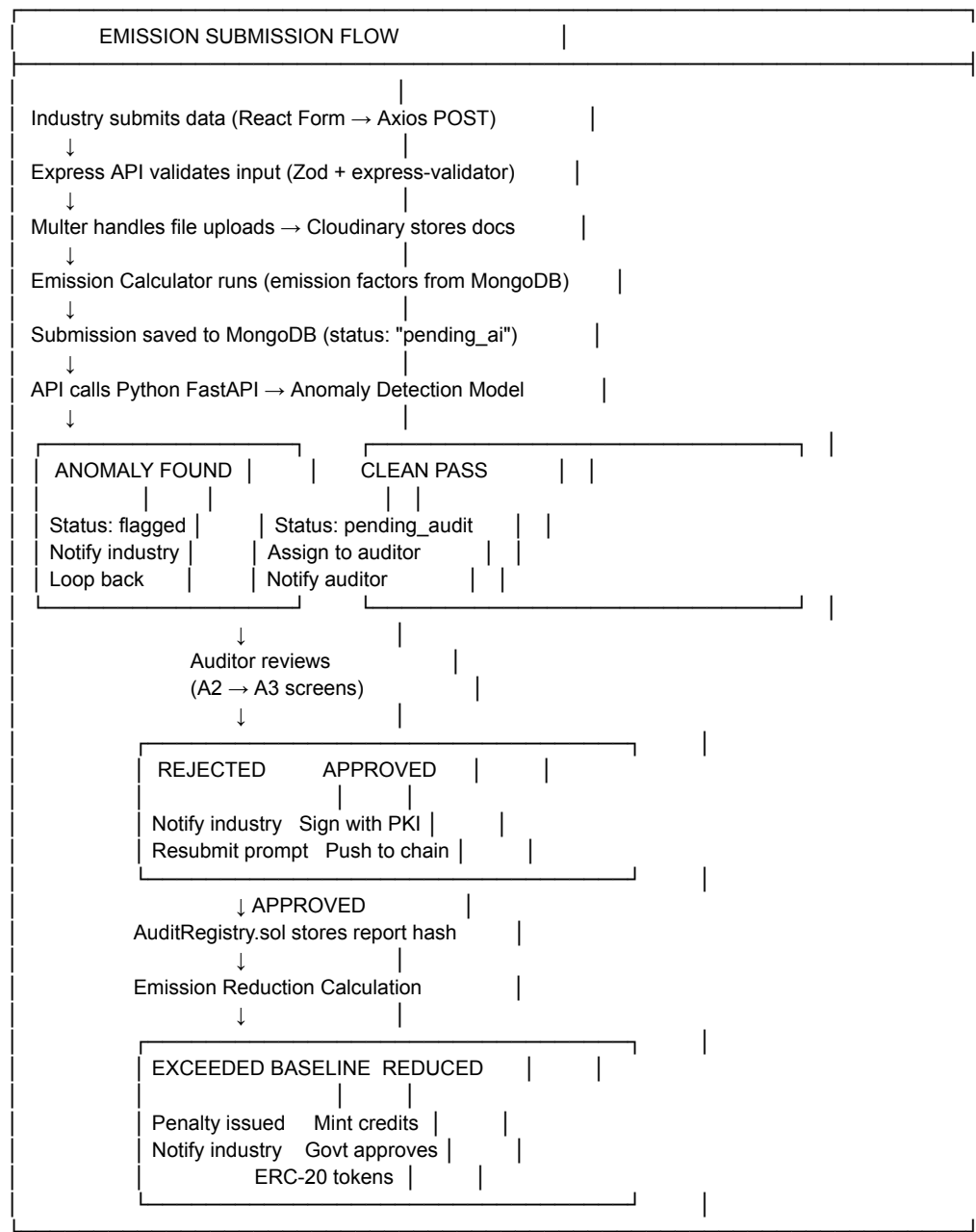




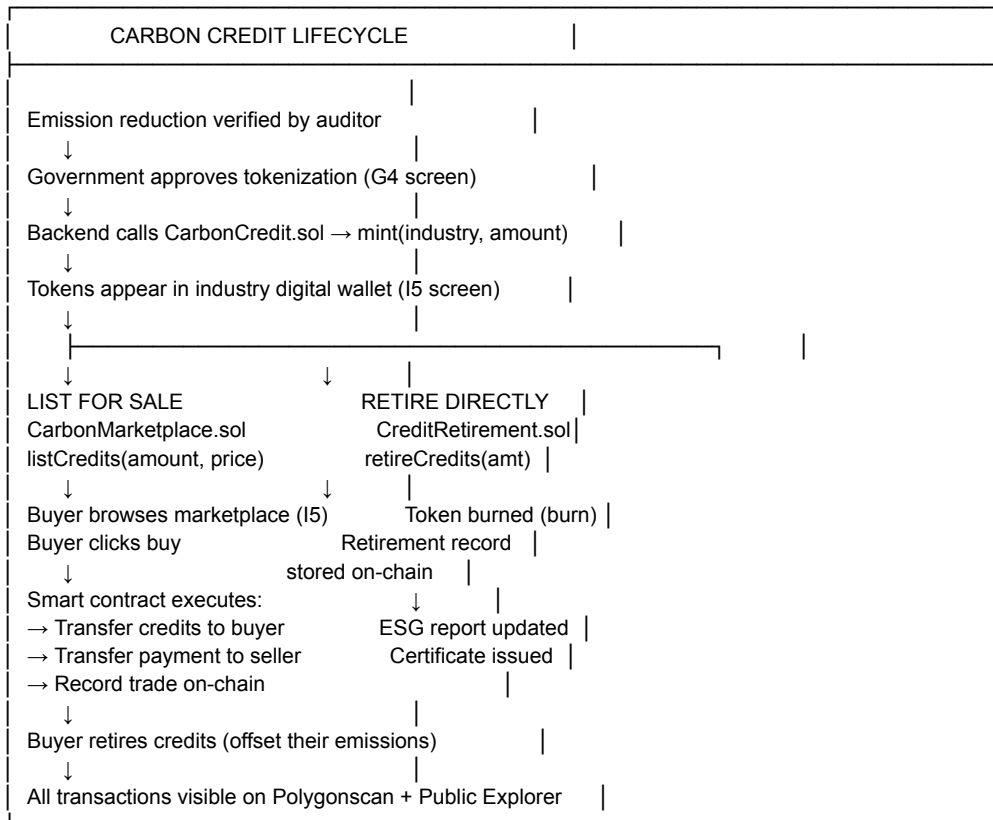
3 Authentication & Authorization Architecture



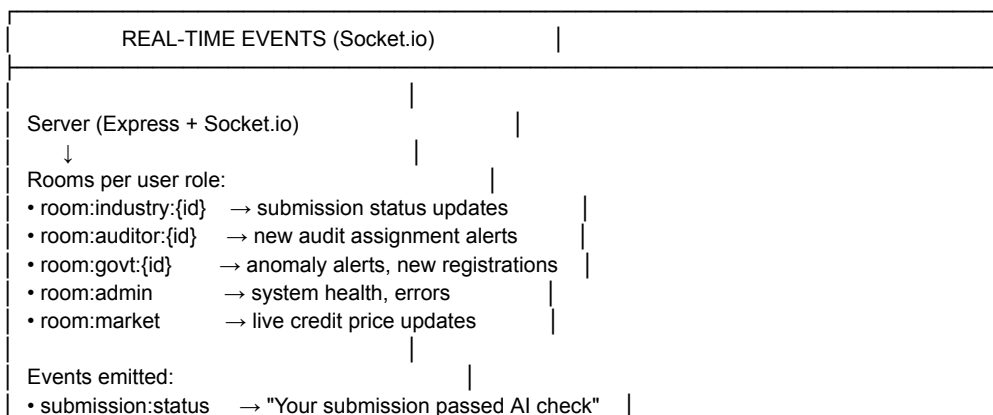
4 Emission Submission Flow Architecture



5 Carbon Credit Flow Architecture

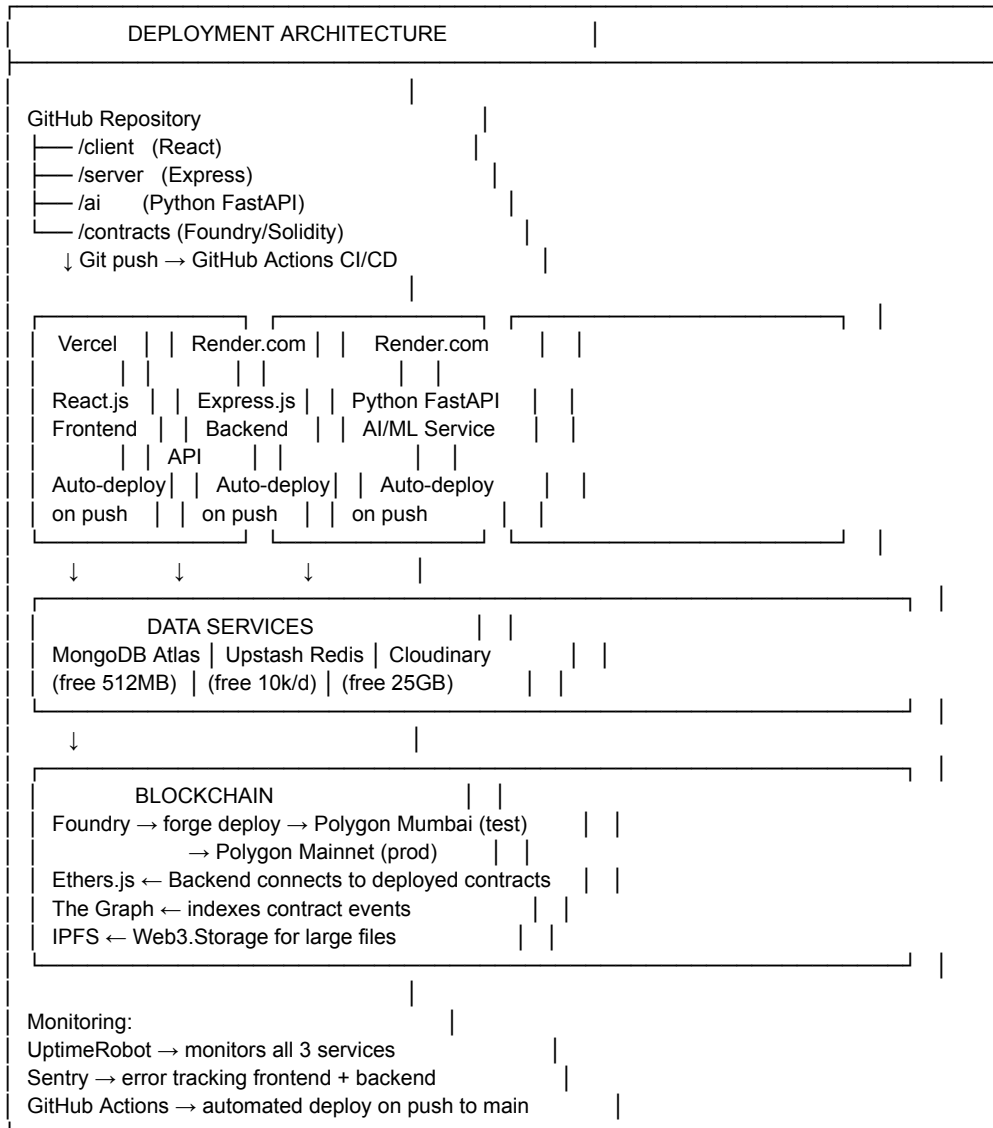


6 Real-time Architecture (Socket.io)



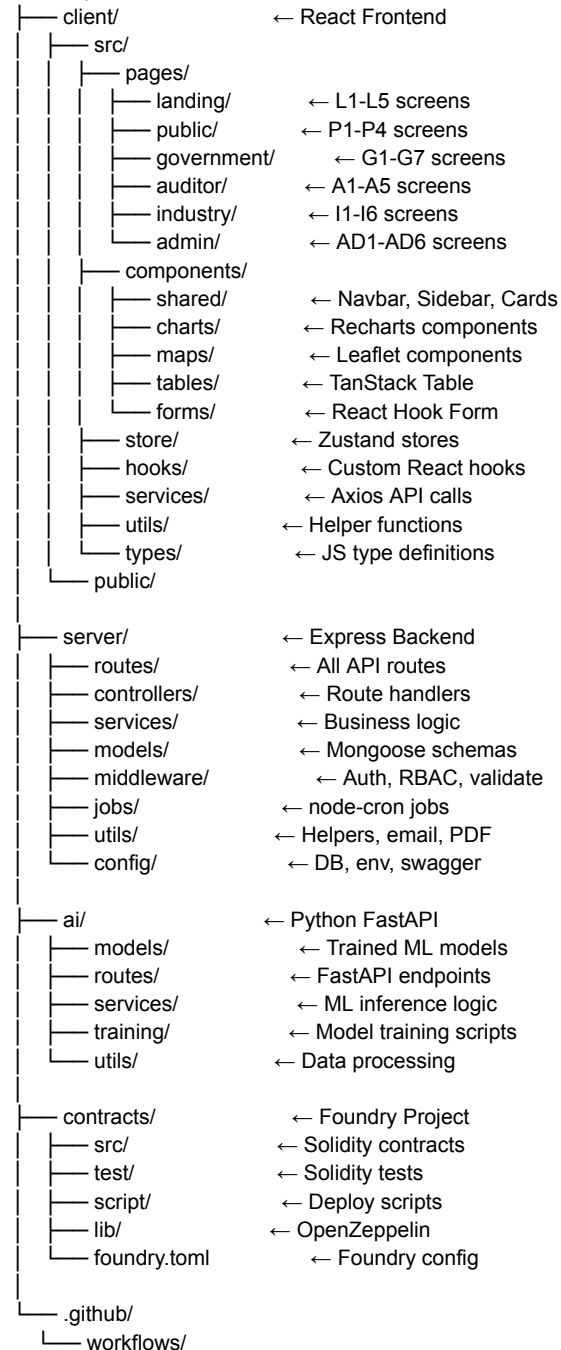
- audit:assigned → "New audit request assigned"
- credit:minted → "10 credits minted to your wallet"
- trade:executed → "Trade completed successfully"
- anomaly:detected → "Anomaly flagged in submission"
- market:priceUpdate → Live credit price change
- system:alert → Admin system notifications

7 Deployment Architecture



8 Folder Structure

carbon-platform/



```

├── deploy-client.yml    ← Vercel deploy
├── deploy-server.yml   ← Render deploy
└── deploy-ai.yml       ← Render AI deploy

```

📊 Architecture Summary

LayerTechnologyHostingCost 🍷 FrontendReact + ViteVercelFree ⚙️ BackendNode + ExpressRenderFree 🗄️ DatabaseMongoDB
 AtlasMongoDB CloudFree 📦 CacheRedisUpstashFree 📁 FilesCloudinaryCloudinaryFree 🚢 AI/MLPython FastAPIRenderFree 🔗
 BlockchainPolygon + FoundryPolygonFree 🌐 CDNVercel EdgeVercelFree 📧 EmailNodemailerGmail SMTPFree 🚧
 RealtimeSocket.ioSelf-hostedFree 📊 MonitoringUptimeRobot + SentryCloudFreeTotal\$0